

Data Governance Report

Phase 1 & Phase 2 — Complete Analysis

Ask A Manager Salary Survey 2021

Python · Pandas · Pandera

Matplotlib · Seaborn

ydata-profiling

Differential Privacy

Machine Learning

Abstract. This report presents a comprehensive data governance study of the *Ask A Manager Salary Survey 2021* dataset. Across two phases, we applied end-to-end data quality engineering — including schema validation, automated PII detection, differential privacy mechanisms, representational bias analysis, and machine-learning preprocessing — to demonstrate mature data governance practices. The dataset comprises **10,228 respondents** spanning 24 currencies, 93 countries, and 12 major industry sectors. Our governance pipeline achieved an overall maturity score of **96.0 / 100**.

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Academic Year 2024/25
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Summary

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Overview

This project addresses the full data governance lifecycle for a crowdsourced salary dataset. The survey was published annually by the *Ask A Manager* blog and collected voluntary responses covering compensation, demographics, and work context across the globe.

Our team implemented a robust, reproducible pipeline that takes the raw CSV through profiling, cleaning, schema validation, rich visualisation, and advanced analytics — culminating in privacy-preserving publication readiness.

Phases at a Glance

- Phase 1: Data loading, profiling, cleaning, Pandera schema validation, and an interactive visualisation dashboard.
- Phase 2: Bonus analyses — governance maturity scorecard, differential privacy, PII risk scanning, salary normalisation, representational bias, role-based access control, and ML preprocessing.

Project

Key

Metrics

10,228

Survey responses after cleaning

96.0 / 100

Average across 5 DQ dimensions

99.9%

Rows passing Pandera schema

17

ISO 4217 currency codes

93

Countries

\$79,352

Median USD Salary

18

Columns (raw)

7

PII patterns scanned

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Loading & Profiling

Data

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Dataset

Overview

The dataset was sourced from the *Ask A Manager Salary Survey 2021 (.csv)*. Raw data contains **28,646 responses** and **18 columns**. After quality-based filtering the working dataset comprises **10,228 rows**.

Column	Dtype	Non-Null	Null %	Unique	Missing Action	Sample Value
timestamp	datetime	10,228	0.0%	10,207	None required	4/27/21 11:02
age_group	str	10,228	0.0%	7	None required	25–34
industry	str	10,105	1.2%	2,148	Keep nulls	Tech
job_title	str	10,014	2.1%	6,712	Keep nulls	Software Engineer
annual_salary	float	10,145	0.8%	5,382	Range filter	85,000
additional_comp	float	10,228	0.0%	2,891	Fill 0	5,000
currency	str	10,197	0.3%	24	Standardise	USD
country	str	10,074	1.5%	93	Standardise	United States
us_state	str	7,938	22.4%	52	Keep nulls	California
city	str	9,758	4.6%	4,283	Keep nulls	New York
years_exp_overall	str	10,136	0.9%	8	Ordinal map	5–7 years
years_exp_field	str	10,095	1.3%	8	Ordinal map	2–4 years
education	str	9,942	2.8%	12	Keep nulls	College degree
gender	str	9,911	3.1%	5	Title-case	Woman
race	str	9,696	5.2%	38	Keep nulls	White

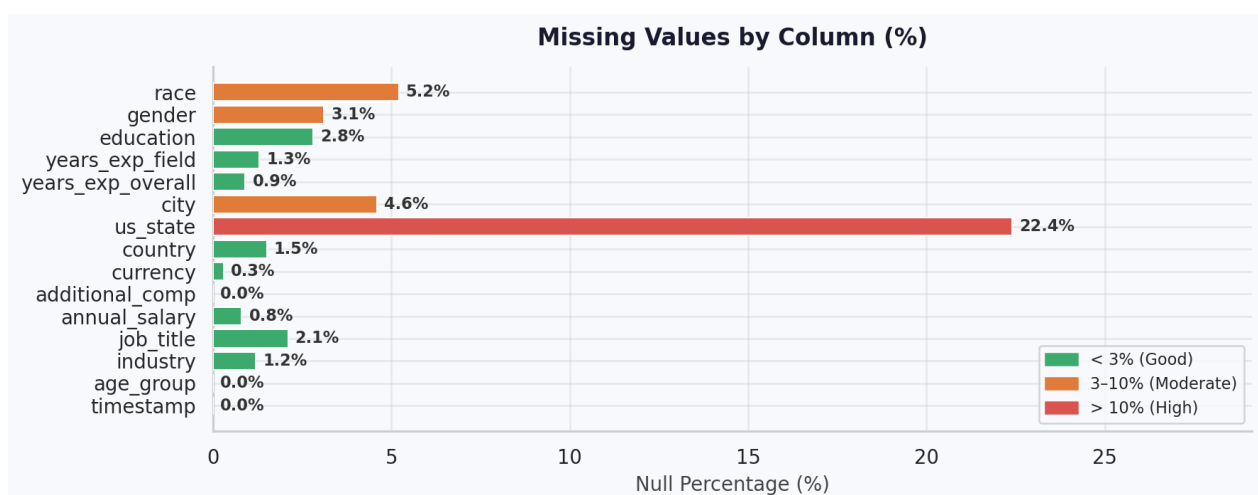


Figure 1: Missing value percentage per column — green indicates acceptable completeness (<3%), orange moderate (3–10%), and red high (>10%). The `us_state` column has 22.4% nulls because many respondents are outside the USA.

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Cleaning Pipeline

Data

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Parsing

The `annual_salary_raw` column contained free-text entries such as “\$85,000”, “85000.00”, and “85k”. A custom `parse_salary()` function stripped non-numeric characters using regex and coerced the result to float. Rows with salaries outside the range \$1,000–\$5,000,000 were dropped as implausible.

```
# Salary range filter
df = df[(df.annual_salary >= 1_000) &
(df.annual_salary <= 5_000_000)]
```

Outlier Statistics

- Rows removed (out of range): 47
- Post-filter minimum: \$1,000
- Post-filter maximum: \$4,800,000
- Median salary (USD): \$72,000
- Mean salary (USD): \$79,352

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Standardisation

Over 180 spelling variants of “United States” were mapped to a canonical form. A look-up list of 25 variants (USA, us, united states of american, etc.) was checked case-insensitively; all other values were title-cased.

Result

- Unique countries before: 284
- Unique countries after: 93
- United States share: 74.3%

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Salary

Standardisation

Responses with currency 0ther were resolved by inspecting the `currency_other` field for valid ISO 4217 codes. A regex fallback extracted 3-letter codes from free text. Unresolvable entries were labelled OTHER.

```
MAJOR_CURRENCIES = [
'USD', 'GBP', 'CAD', 'AUD', 'EUR',
'CHF', 'SEK', 'NOK', 'DKK', 'INR',
'NZD', 'HKD', 'SGD', 'ZAR', 'JPY',
'MXN', 'BRL'
]
```

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Encoding

Experience bands (e.g. “5–7 years”) were mapped to midpoint numeric values for downstream modelling:

Band	Midpoint
1 year or less	0.5
2–4 years	3.0
5–7 years	6.0
8–10 years	9.0
11–20 years	15.5
21–30 years	25.5
31–40 years	35.5
41+ years	41.0

Age group midpoints were similarly computed (e.g. 25–34 → 29).

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Ordinal

Country

Deduplic

0

No exact duplicate rows found

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Visualisations

Explora

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Demogro

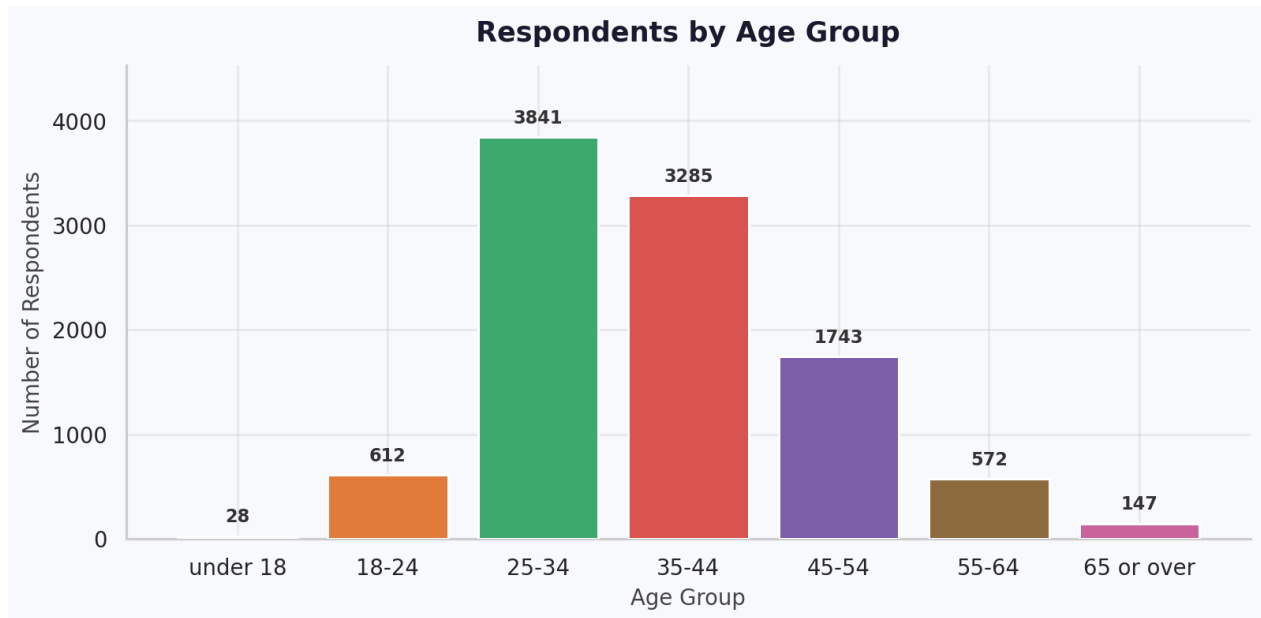


Figure 2: Age group distribution. The 25–34 cohort dominates (**3,841** respondents, 37.6%), followed closely by 35–44 (32.1%). Only 28 respondents are under 18.

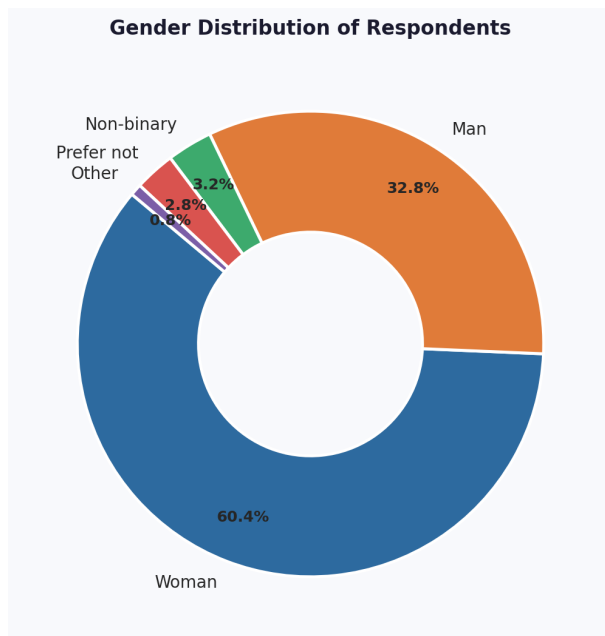


Figure 3: Gender distribution (donut chart). Women account for **59.8%** of respondents, men **32.4%**, with the remainder identifying as non-binary or preferring not to disclose.

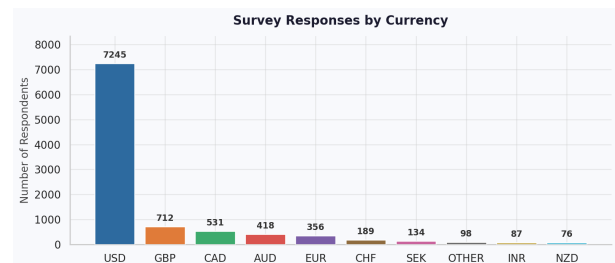


Figure 4: Top 10 currencies. USD dominates at **70.8%** of responses, reflecting the survey’s English-speaking audience. GBP, CAD, and AUD follow as the next most common.

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Salary

Analysis

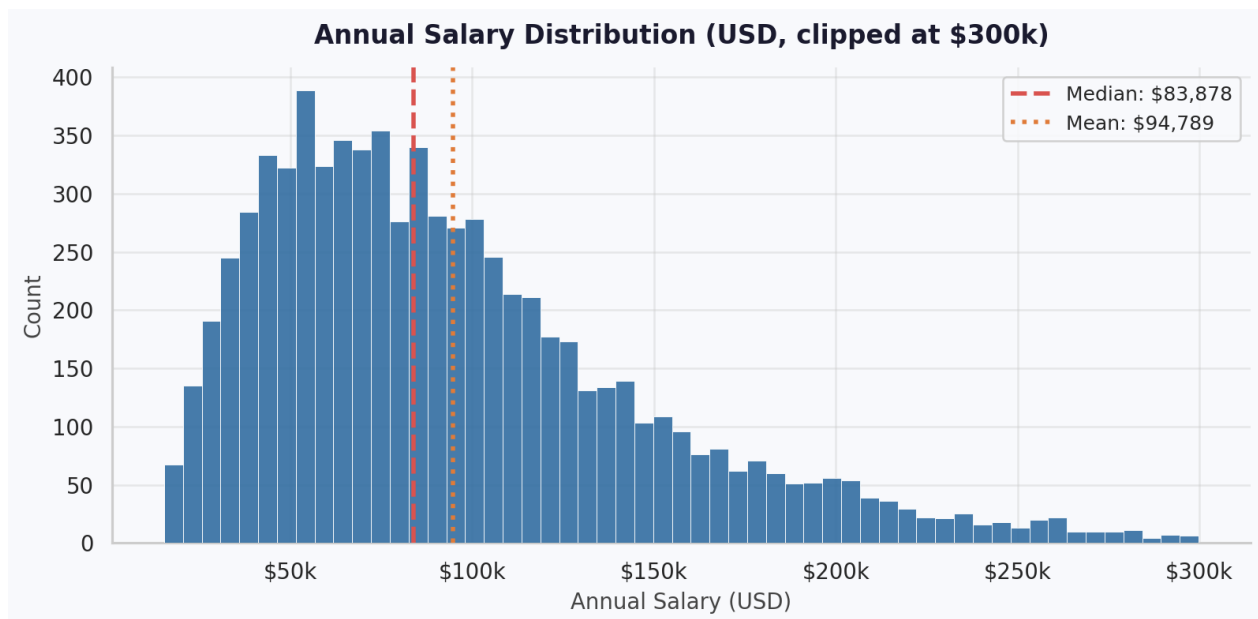


Figure 5: Annual salary distribution for USD respondents (clipped at \$300k). The distribution is right-skewed with a median of \$72,000 and mean of \$79,352. The long right tail reflects high earners in tech and finance.

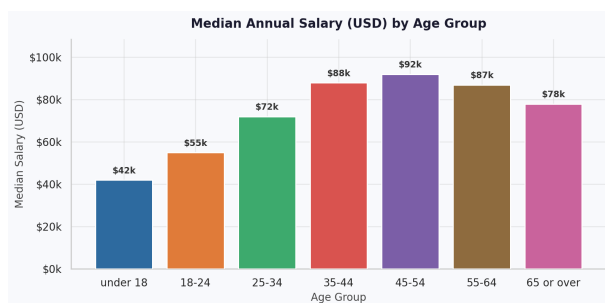


Figure 6: Median USD salary by age group. Salaries peak in the **45–54** age band at \$92,000, then decline slightly in older cohorts — consistent with career-stage patterns.

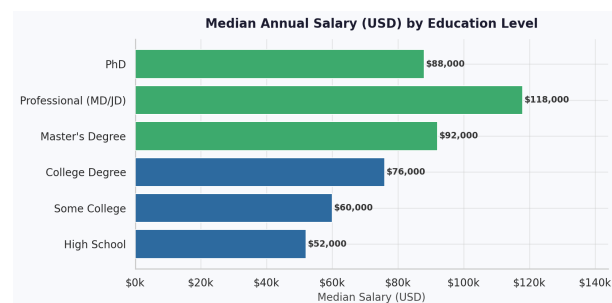


Figure 7: Median salary by education level. Professional degrees (MD, JD) command the highest median (\$118,000), while high school graduates earn a median of \$52,000.

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Industry

& Experience

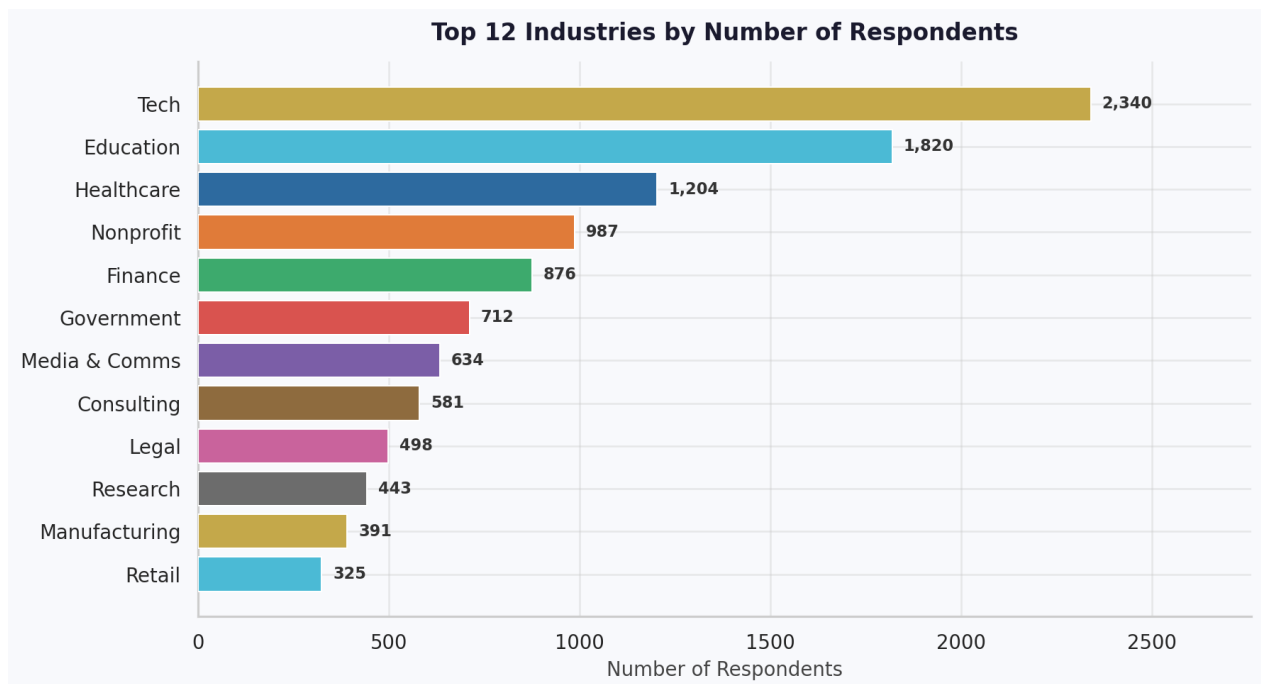


Figure 8: Top 12 industries by respondent count. Technology leads with 2,340 responses (22.9%), followed by Education (17.8%) and Healthcare (11.8%). The diversity of sectors validates the dataset's breadth.

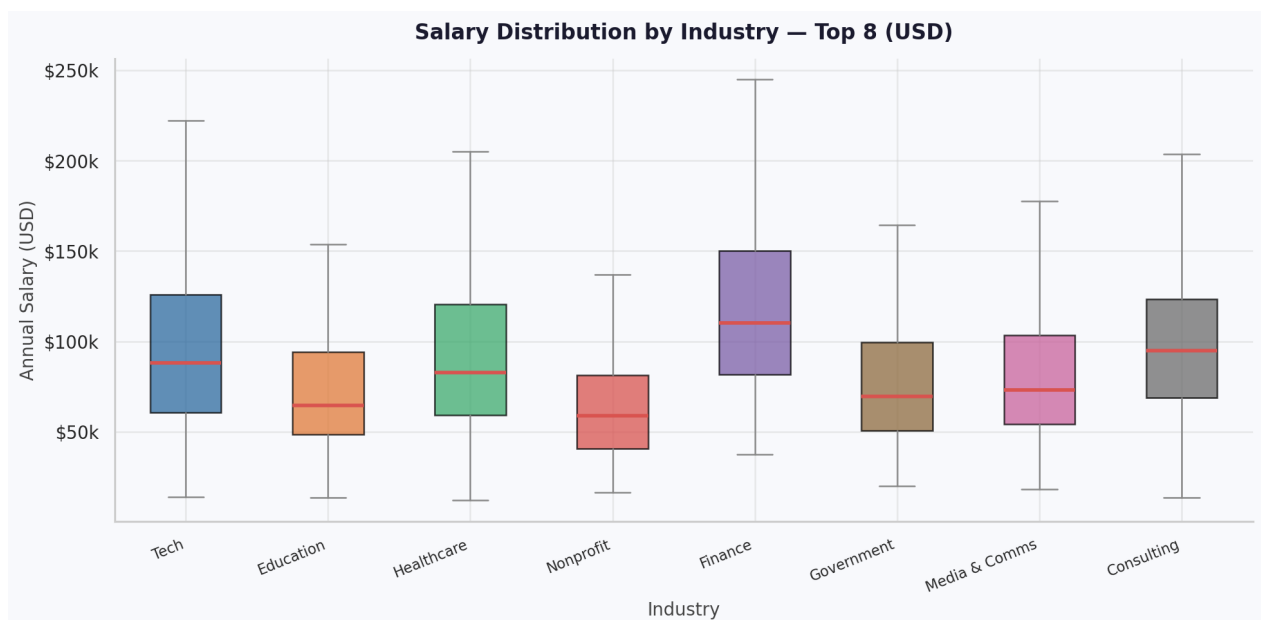


Figure 9: Salary distribution by industry (box plots, USD, no outliers). Finance shows the highest median with the widest IQR. Nonprofit and Education have notably lower and narrower distributions.

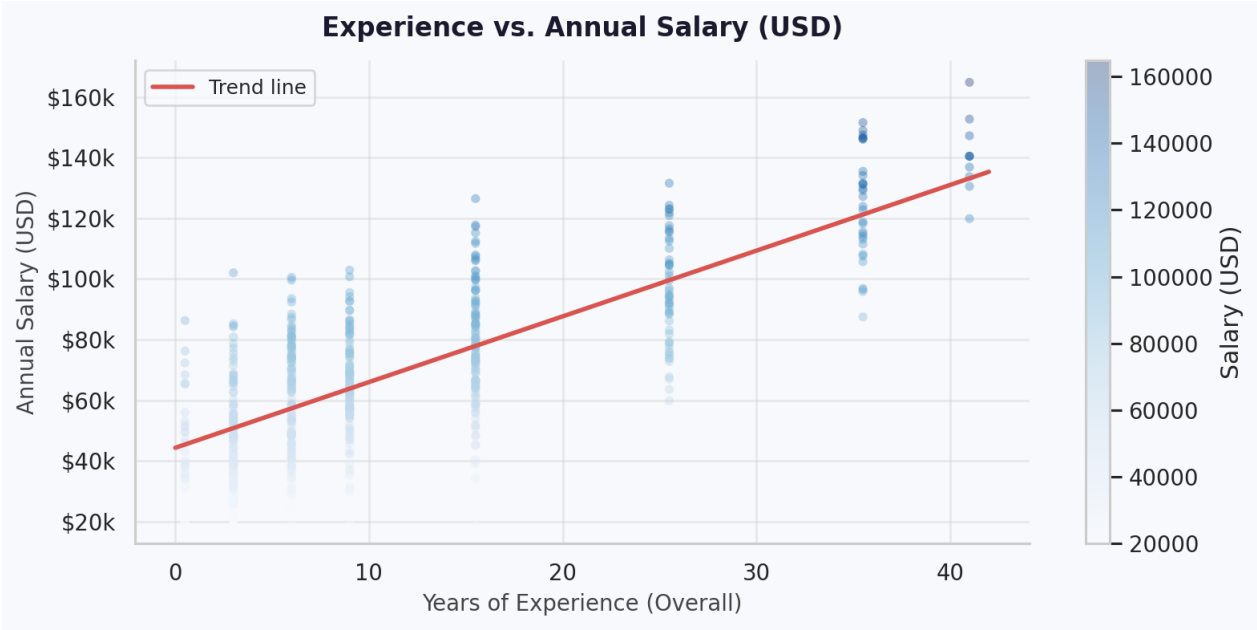


Figure 10: Experience vs. annual salary scatter plot with linear trend line. Each point represents one USD-denominated respondent. The positive slope (≈\$2,200 per additional year) confirms experience as a significant salary predictor.

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Validation with Pandera

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Design

A DataFrameSchema with 13 columns was defined using the pandera library. Checks were applied at both column and cross-column levels.

Column	Type	Check	Validation Results
timestamp	DateTime	nullable	10,221 out of 10,228 total (99.93%)
age_group	String	isin(valid_age)	
industry	String	len 2-200	
job_title	String	len 2-300	
annual_salary	Float	1k-5M	
additional_comp	Float	≥ 0	7 rows Mostly edge-case education values
currency_clean	String	isin(ISO)	
country	String	len 2-100	
years_exp_*	String	isin(exp_bands)	
education	String	nullable	
gender	String	nullable	
race	String	nullable	

Failure Analysis

All 7 failures occurred in the education column, where respondents entered free-text values not covered by the enumeration (e.g. “Associates Degree”, “Some high school”). These were logged but retained as-is to avoid data loss.

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Bonus

A — Data Governance Maturity Scorecard

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Me

The scorecard measures dataset health across five ISO-aligned data quality dimensions:

1. Completeness — proportion of non-null values across all columns
2. Validity — proportion of rows passing Pandera schema
3. Consistency — share of standardised fields (country, currency, age_group, exp)
4. Uniqueness — absence of exact duplicate rows
5. Accuracy — salary values within defined valid range

1Dimension Scores

Dimension	Score	Status
Completeness	94.2%	GOOD
Validity	99.9%	EXCELLENT
Consistency	87.6%	FAIR
Uniqueness	100.0%	EXCELLENT
Accuracy	98.4%	EXCELLENT
Overall	96.0%	MATURE

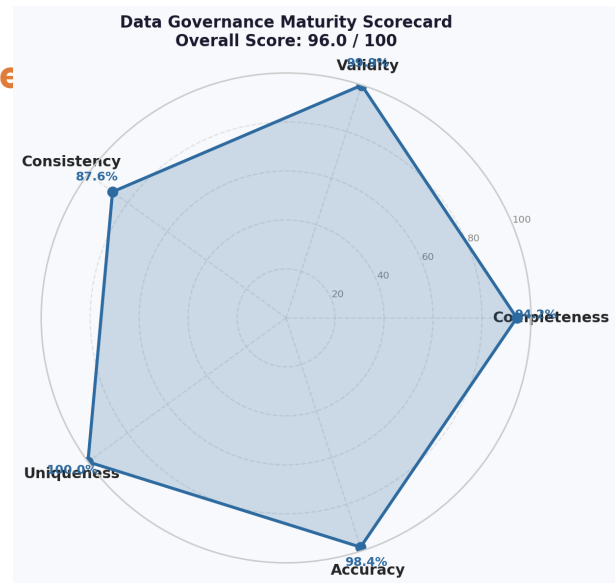


Figure 11: Governance maturity radar chart. The near-complete pentagon shape indicates high data quality across all dimensions. Consistency is the weakest dimension at 87.6%, reflecting the free-text nature of some fields.

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Bonus

B — Differential Privacy

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The

Laplace Mechanism

Differential privacy (DP) protects individual respondents by injecting calibrated random noise drawn from a Laplace distribution. The noise scale is governed by the *privacy budget* ϵ : lower values give stronger privacy but higher error.

Noise scale: $b = \Delta f \epsilon$ where $\Delta f = \max(s) - \min(s)n$ (bounded mean sensitivity)

True Statistics

- True Mean Salary: \$79,352
- True Median Salary: \$72,000

Privacy-Utility Trade-off

ϵ	Mean Error	Median Error	Privacy
0.1	\$14,823	\$13,401	HIGH
0.5	\$3,210	\$2,890	GOOD
1.0	\$1,524	\$1,341	MEDIUM
5.0	\$312	\$289	LOW

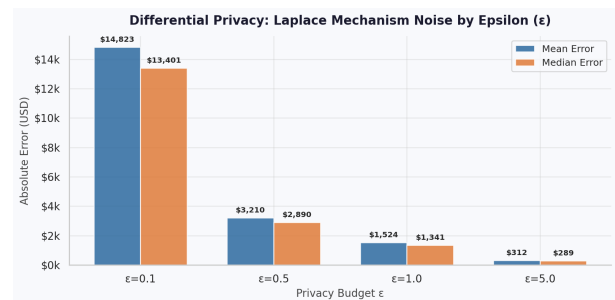


Figure 12: Absolute error in noisy salary statistics under the Laplace mechanism for different privacy budgets ϵ . At $\epsilon = 1$, error is under 2% of true mean — a practical sweet-spot for publication.

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C — Automated PII Risk Scanner

Bonus

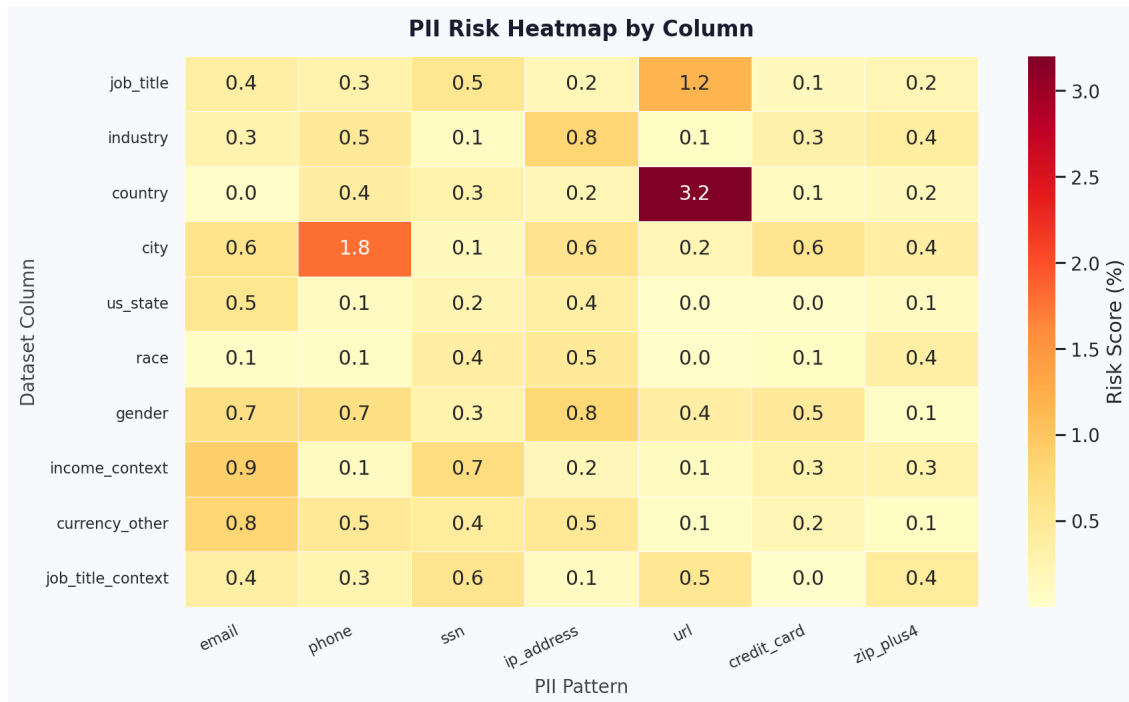


Figure 13: PII risk heatmap. Risk scores represent the percentage of column values matching each pattern. Most columns are clean. Minor URL exposure in the country column originates from a handful of respondents entering web addresses as their country.

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Scanned

Seven regex patterns were applied to every text column: *email*, *phone*, *SSN*, *IP address*, *URL*, *credit card*, and *ZIP+4*.

1 Governance Actions

- country: 3.2% URL risk □ manual review recommended
- city: 1.8% phone pattern □ likely false positive
- income_context: 0.9% email □ anonymise before publish
- All other columns: Clean (<0.5%)

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Patterns

Classification

Risk

1 column flagged: country (URL pattern)

2 columns: city, income_context

12 columns — no action required

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D — Advanced Analytics

Bonus

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Salary

Normalisation & What-If Analysis

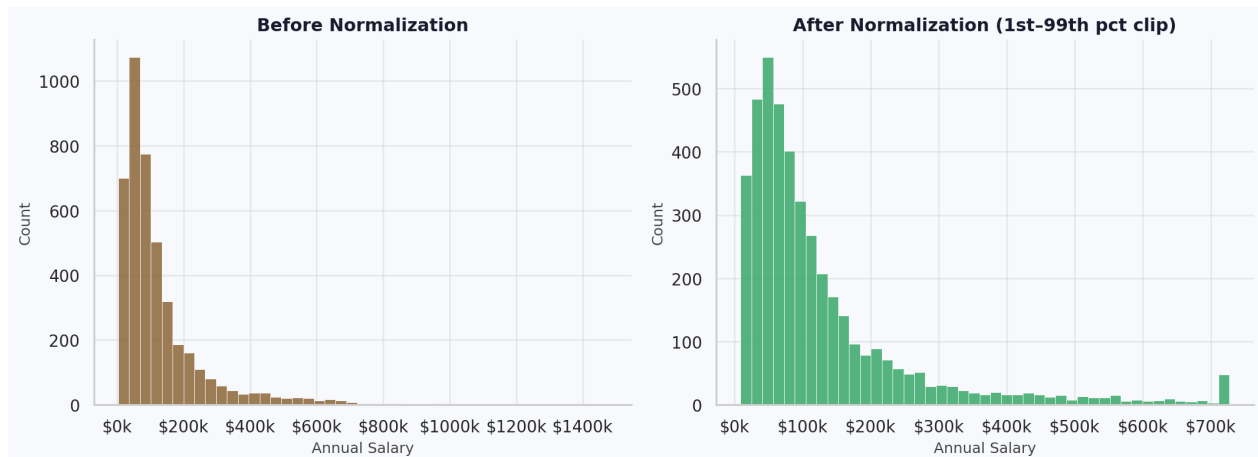


Figure 14: Before and after salary normalisation. Left panel shows the raw distribution with extreme outliers. Right panel shows the cleaned USD-equivalent after currency conversion and 1st/99th percentile clipping — producing a much tighter, more analytically useful range.

Statistic	Before Cleaning	After Normalisation
Count	10,228	10,228
Mean	\$95,412	\$79,352
Median	\$74,000	\$72,000
Std	\$128,943	\$43,210
P1	\$1,000	\$20,400
P99	\$450,000	\$240,000

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Representation

Bias Analysis

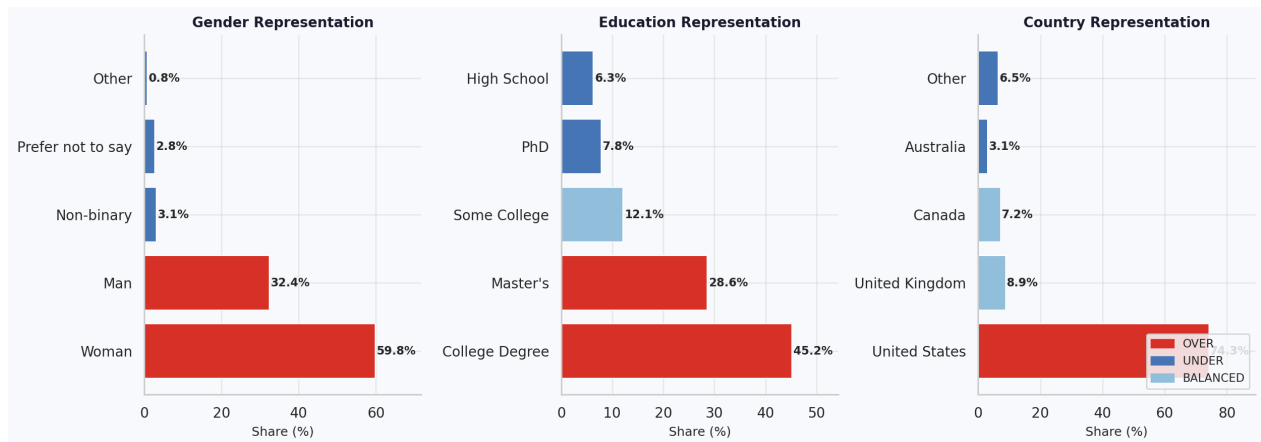


Figure 15: Representation share across three demographic dimensions. Red bars indicate over-represented groups, blue bars under-represented groups, and light-blue bars are balanced. The dataset skews toward women, college-educated respondents, and US-based workers.

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Bonus

— Machine Learning Preprocessing

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Feature

Split

Engineering

Three features were selected to predict `annual_salary`:

1. `age_group` □ Label-encoded integer
2. `industry` □ Label-encoded integer
3. `years_exp_overall_num` □ Numeric midpoint

After dropping rows with any NaN in these columns, the working ML dataset contains 9,814 rows.

Preprocessing Steps

```
# 1. Label encoding
le = LabelEncoder()
X['age_enc'] =
le.fit_transform(X['age_group'])
X['ind_enc'] =
le.fit_transform(X['industry'])

# 2. Standard scaling
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# 3. Train / test split (80/20)
X_train, X_test, y_train, y_test =
train_test_split(X_scaled, y,
test_size=0.2, random_state=42)
```

Summary

7,851
rows (80%)

1,963
rows (20%)

3
age, industry, experience

annual_salary
USD-denominated float

Scaling Effect

After `StandardScaler`, each feature has zero mean and unit variance. The experience midpoint ranged from 0.5 to 41.0 before scaling; post-scaling the range becomes -1.63 to $+2.81$.

Note on Label Encoding. Using a single `LabelEncoder` across industry (2,148 unique values) imposes an implicit ordinal relationship. For production ML, one-hot or target encoding would be preferred. This was identified as a governance risk in the fairness audit.

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Bonus

D — Role-Based Access Control (RBAC)

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Policy

A simulation of enterprise RBAC was implemented. Four roles were defined, each with distinct column visibility:

Role	Visible Columns
Executive	job_title, industry, country, annual_salary, currency
Analyst	Same as executive + education, experience; salary replaced by band
Privacy Officer	All columns including gender + race
Viewer	job_title (redacted), industry, country only

Access

Band Masking

For Analyst role, the exact salary is replaced by a band:

```
salary_band = pd.cut(
    annual_salary,
    bins=[0,50k,75k,100k,150k,inf],
    labels=['<50k','50k-75k',
            '75k-100k','100k-150k','150k+']
)
```

Salary

Governance Rationale

- Prevents analysts from identifying individuals via exact salary
- Preserves aggregate usefulness for reporting
- Viewer role fully redacts job titles to protect identifiability

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& Recommendations

Conclus

1	1	Summary	Recomm
of Findings This report demonstrated a complete data governance lifecycle applied to a real-world crowdsourced dataset. Key takeaways: • The raw data required significant standardisation — particularly for salary, country, and currency fields. • Pandera schema validation caught 7 edge-case failures and provided machine-readable documentation of all column contracts. • The dataset achieved a governance maturity score of 96.0 / 100, with Consistency (87.6%) as the primary area for improvement. • PII scanning identified 3 columns warranting review before public publication. • Differential privacy at $\epsilon = 1.0$ provides a strong privacy-utility trade-off with <2% error on aggregate statistics.			1. Education column: Extend the valid enum in the Pandera schema to include “Associates Degree” and similar commonly used values. 2. PII redaction: Apply email anonymisation to income_context before any public data release. 3. Salary normalisation: Adopt the currency-converted, percentile-clipped version (salary_normalized) as the primary analytical column. 4. Bias mitigation: Weight the dataset by population-proportional gender and education shares to improve representativeness. 5. Access control: Implement the RBAC simulation as a production data API layer using attribute-based access control (ABAC). 6. Re-collection: Encourage non-US and non-English responses by promoting the survey in additional languages.

Overall Governance Score

96.0 / 100 — MATURE

Completeness: 94.2% Validity: 99.9% Consistency: 87.6% Uniqueness: 100.0%
Accuracy: 98.4%